# FIBRE PRODUCTS, EQUALIZERS AND FINAL OBJECTS IN THE BIG AND SMALL PRO-ETALE SITES AGREE WITH THOSE OF SCHEMES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the identification of the fibre products through the closure properties of the site and the stability of weak etaleness and affineness, and the explicit description of the equalizer as a fibre product over a diagonal together with the verification that it remains weakly etale.

*Editorial note.* The label lemma-fibre-products-proetale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-fibre-products-proetale –start 2549 –end 2604 –local-id 3584 returned RESULT=OK (L3584 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Closure properties defining the big pro-etale site. Sets, the lemma describing which schemes belong to a site constructed from a function bounding cardinalities. Supplies the membership of the products, fibre products and equalizers in the big pro-etale site.
- Weakly etale morphisms are stable under fibre products and equalizers. More on Morphisms, the section on weakly etale morphisms, in particular their stability under base change and the corresponding closure properties. Supplies the fact that fibre products and equalizers of weakly etale morphisms over S are again weakly etale over S, which gives the fibre products and equalizers in the small pro-etale site.

*Source reference.* Original section 094N: sources/stacks-chow/tags.txt, source lines 11556–11556. Prerequisite pointer invoked at proof line(s) 2580; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 000R: sources/stacks-chow/tags.txt, source lines 36–36. Prerequisite pointer invoked at proof line(s) 2569; it names a standard fact

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of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $S$ be a scheme. Let $Sch_{pro\text{-}étale}$ be a big pro-étale site containing $S$. Let $Sch$ be the category of all schemes.*

1. *The categories $Sch_{pro\text{-}étale}$, $(Sch/S)_{pro\text{-}étale}$, $S_{pro\text{-}étale}$, and $(Aff/S)_{pro\text{-}étale}$ have fibre products agreeing with fibre products in $Sch$.*
2. *The categories $Sch_{pro\text{-}étale}$, $(Sch/S)_{pro\text{-}étale}$, $S_{pro\text{-}étale}$ have equalizers agreeing with equalizers in $Sch$.*
3. *The categories $(Sch/S)_{pro\text{-}étale}$, and $S_{pro\text{-}étale}$ both have a final object, namely $S/S$.*
4. *The category $Sch_{pro\text{-}étale}$ has a final object agreeing with the final object of $Sch$, namely $\mathrm{Spec}(\mathbf{Z})$.*

*Proof.* The category $Sch_{pro\text{-}étale}$ contains $\mathrm{Spec}(\mathbf{Z})$ and is closed under products and fibre products by construction, see Sets, Lemma 000R. Suppose we have $U \to S$, $V \to U$, $W \to U$ morphisms of schemes with $U, V, W \in \mathrm{Ob}(Sch_{pro\text{-}étale})$. The fibre product $V \times_U W$ in $Sch_{pro\text{-}étale}$ is a fibre product in $Sch$ and is the fibre product of $V/S$ with $W/S$ over $U/S$ in the category of all schemes over $S$, and hence also a fibre product in $(Sch/S)_{pro\text{-}étale}$. This proves the result for $(Sch/S)_{pro\text{-}étale}$. If $U \to S$, $V \to U$ and $W \to U$ are weakly étale then so is $V \times_U W \to S$ (see More on Morphisms, Section 094N) and hence we get fibre products for $S_{pro\text{-}étale}$. If $U, V, W$ are affine, so is $V \times_U W$ and hence we get fibre products for $(Aff/S)_{pro\text{-}étale}$.

Let $a, b : U \to V$ be two morphisms in $Sch_{pro\text{-}étale}$. In this case the equalizer of $a$ and $b$ (in the category of schemes) is

$$V \times_{\Delta_{V/\mathrm{Spec}(\mathbf{Z})}, V \times_{\mathrm{Spec}(\mathbf{Z})} V, (a,b)} (U \times_{\mathrm{Spec}(\mathbf{Z})} U)$$

which is an object of $Sch_{pro\text{-}étale}$ by what we saw above. Thus $Sch_{pro\text{-}étale}$ has equalizers. If $a$ and $b$ are morphisms over $S$, then the equalizer (in the category of schemes) is also given by

$$V \times_{\Delta_{V/S}, V \times_S V, (a,b)} (U \times_S U)$$

hence we see that $(Sch/S)_{pro\text{-}étale}$ has equalizers. Moreover, if $U$ and $V$ are weakly-étale over $S$, then so is the equalizer above as a fibre product of schemes weakly étale over $S$. Thus $S_{pro\text{-}étale}$ has equalizers. The statements on final objects is clear. □